

A Sharp Threshold Atlas for Irreducible Network SIS: Global Endemicity and the Critical Perron Law

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Abstract

We close the full phase portrait of the deterministic susceptible–infected–susceptible flow on an irreducible heterogeneous network. The spectral abscissa of the linearized Metzler matrix separates extinction from a unique endemic state. Above threshold, strict subhomogeneity gives global attraction, a Hurwitz endemic Jacobian, and componentwise transmission-rate monotonicity. No epidemic object is promoted to an arithmetic orbit or target determinant.

1 Frozen network flow

Let $A \geq 0$ be irreducible, let $D = \text{diag}(\delta_i)$ with $\delta_i > 0$, and let $\beta > 0$. On $Q = [0, 1]^n$ consider

$$\dot{x} = \beta \text{diag}(1 - x)Ax - Dx, \quad M = \beta A - D. \quad (1)$$

This is the physical ODE clock. Lajmanovich and Yorke established the classical nonhomogeneous deterministic epidemic lineage [1]; all normalizations below are stated locally.

At $x_i = 0$, the i th component of the vector field is nonnegative; at $x_i = 1$ it equals $-\delta_i$. Thus Q is invariant. Irreducibility and cooperativity make every nonzero orbit strictly positive for positive time.

Theorem 1 (threshold theorem). *If $s(M) < 0$, zero is globally exponentially stable. If $s(M) = 0$, zero is globally asymptotically stable. If $s(M) > 0$, (1) has exactly one $x_* \in (0, 1)^n$, and x_* attracts every nonzero point of Q .*

Proof. For $s = s(M) \leq 0$, choose $w \gg 0$ with $w^T M = sw^T$. Then

$$\frac{d}{dt} w^T x = sw^T x - \beta w^T \text{diag}(x)Ax.$$

This proves exponential decay for $s < 0$ and, by positivity and LaSalle’s argument, convergence to zero at equality. For $s > 0$, a small Perron vector is a strict subsolution and $\mathbf{1}$ is a strict supersolution. Existence follows by monotone iteration. On positive states,

$$f(\theta x) - \theta f(x) = \beta \theta (1 - \theta) \text{diag}(x)Ax \gg 0, \quad 0 < \theta < 1. \quad (2)$$

Largest- and smallest-ratio comparisons use (2) to prove uniqueness and to squeeze every positive orbit to x_* . $\square$

2 Endemic stability and monotonicity

At the endemic state,

$$J_* = \beta \text{diag}(1 - x_*)A - \beta \text{diag}(Ax_*) - D. \quad (3)$$

It is irreducible Metzler and $J_* x_* = -\beta \text{diag}(Ax_*)x_* \ll 0$; hence J_* is Hurwitz. Differentiating the equilibrium equation gives

$$\frac{dx_*}{d\beta} = -J_*^{-1} \text{diag}(1 - x_*)Ax_* \gg 0, \quad (4)$$

because $-J_*^{-1}$ is strictly positive. Equations (3)–(4) close both the local spectrum and the parameter direction, rather than stopping at an existence threshold.

References

- [1] A. Lajmanovich and J. A. Yorke, *A deterministic model for gonorrhea in a nonhomogeneous population*, Mathematical Biosciences 28 (1976), 221–236, [doi:10.1016/0025-5564\(76\)90125-5](https://doi.org/10.1016/0025-5564(76)90125-5).